

Hidden Gems of Everyday Game Development with C#

Andrea Magnorsky
@silverspoon

roundcrisis.com

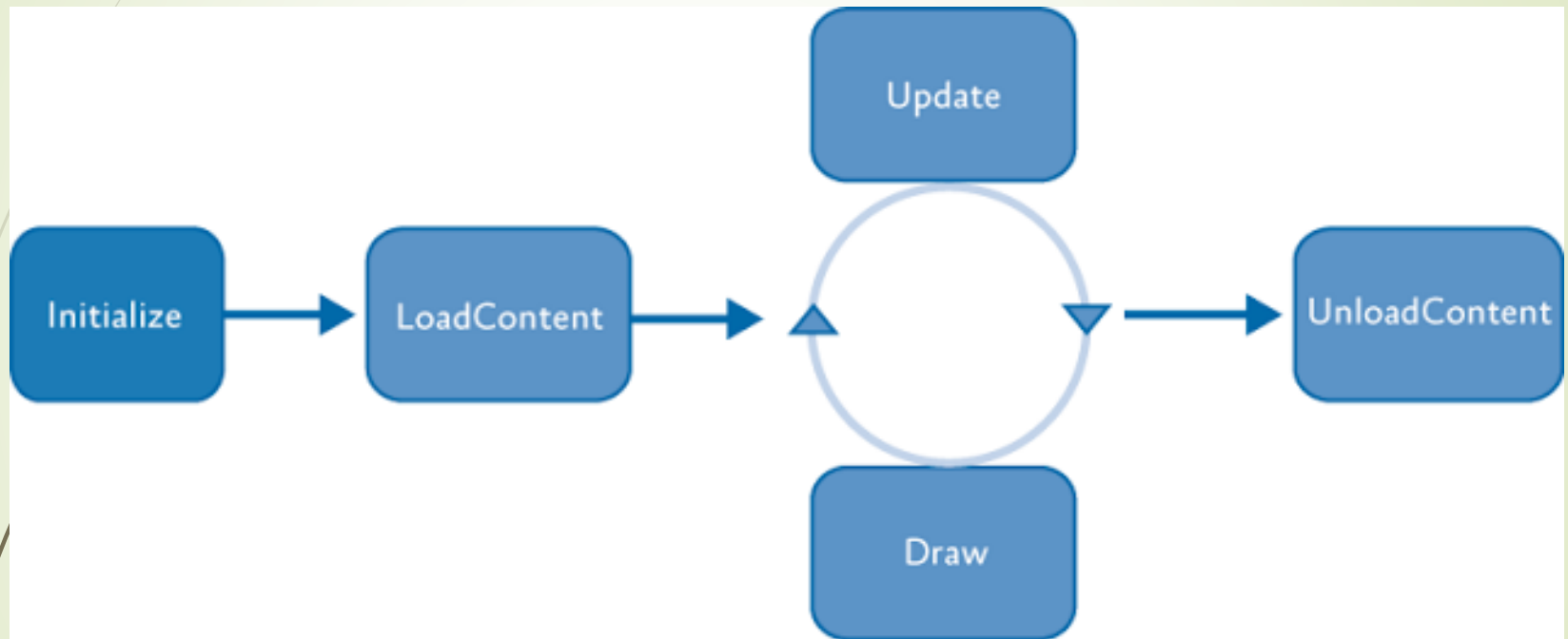


The image is a title screen for the game 'Honourbound'. It features a dramatic sunset or sunrise scene with a bright, glowing sun low on the horizon, casting a warm orange and yellow light across the sky and the landscape. The sky is filled with dark, swirling clouds. In the foreground, a samurai warrior is silhouetted against the bright light, standing on a dark, grassy field. The warrior is wearing traditional armor and holding a long sword. Several tall, thin, rectangular banners or flags are planted in the ground, some of which are also silhouetted against the light. In the background, there are rolling hills or mountains. The overall mood is epic and atmospheric.

H O N O U R B O U N D

PRESS START

The humble Game Loop



Hello Dovahkiin



34 responses

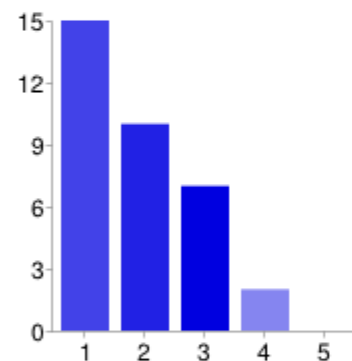
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Summary

What type of game are you developing?

Action-Platformer 2D space shooter wargame Retro games Ending runner Tile based 2D top down space sandbox learning project A bada
PC. mobile "runner" game Simulation Edutainment tower defense Strategy/simulation Social Puzzle Sidescroller Platformer Vehicular
Continuous runner asdas MMORPG first-person RPG RTS RougeLike Single screen shoot 'em up Action adventure 2D platformer

How "big" is the game you are developing?



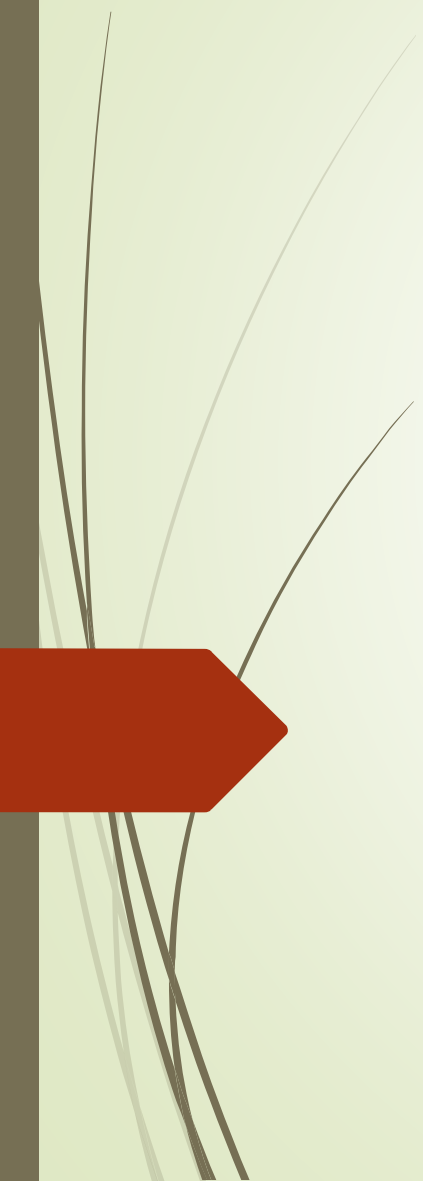
1	15	44%
2	10	29%
3	7	21%
4	2	6%
5	0	0%

What are the top 3 programming languages you use to write games

Python, C++ C++ / AS3 EIFFEL C++ C C++, C#, Javascript C#, R6RS Scheme C#, Actionscript, JavaScript C++ Python Lua Java Pyt
Maker's script thing C++, C#, Lua, Python (sorry, i have a hard time counting) Java YAML GLSL F#, C#, Lua C++ JavaScript Java F# C# C
JavaScript Haskell (Main language) Reactive-banana (Domain Specific Language inside Haskell) C (For parts that require performance, such as path fir
used Lua to script games in a game creation based game called roblox. Im making the game mentioned in the first answer using C++. As a third language
actual languages than Lua and C++ Java C# Python I guess? Mostly just those first two though :) C#, UnityScript, java Scripting Language Java C
exclusively. I have implemented scripting support through IronPython, but haven't really used it. C++ C# Java c# C++

Delegates, func, etc



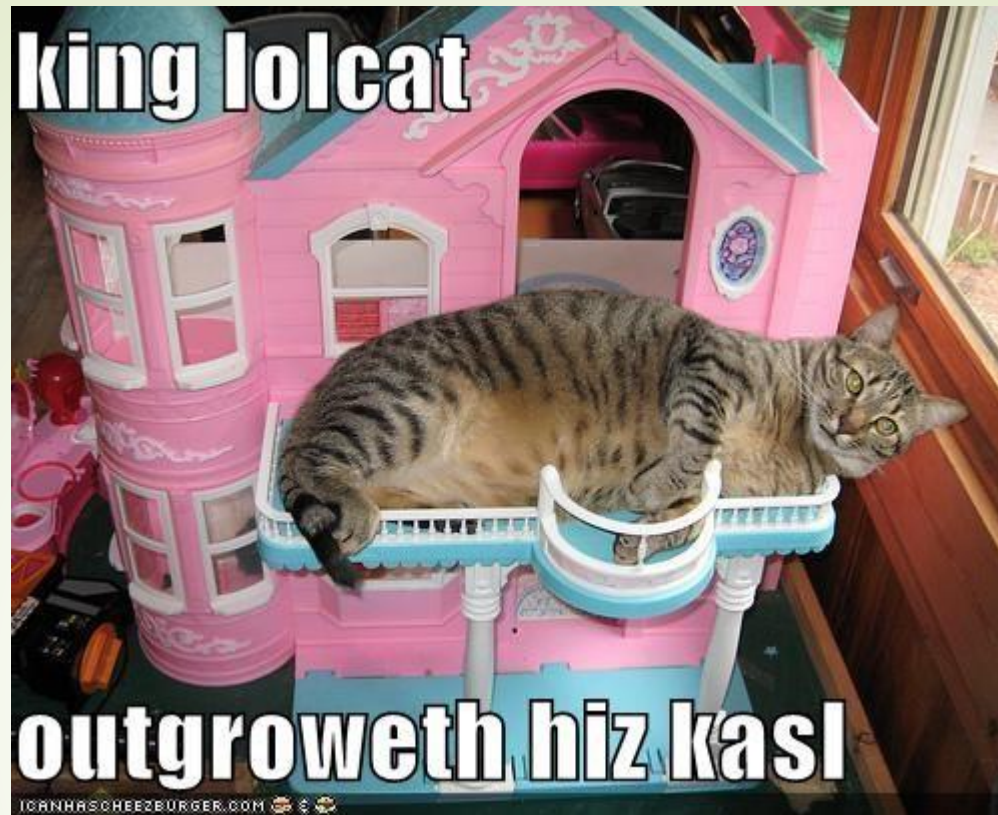


Don't use `linq`



Coroutines

“Coroutines are computer program components that generalize subroutines to allow multiple entry points for suspending and resuming execution at certain locations”



Why?

➡ _changePositionCoroutine.MoveNext()

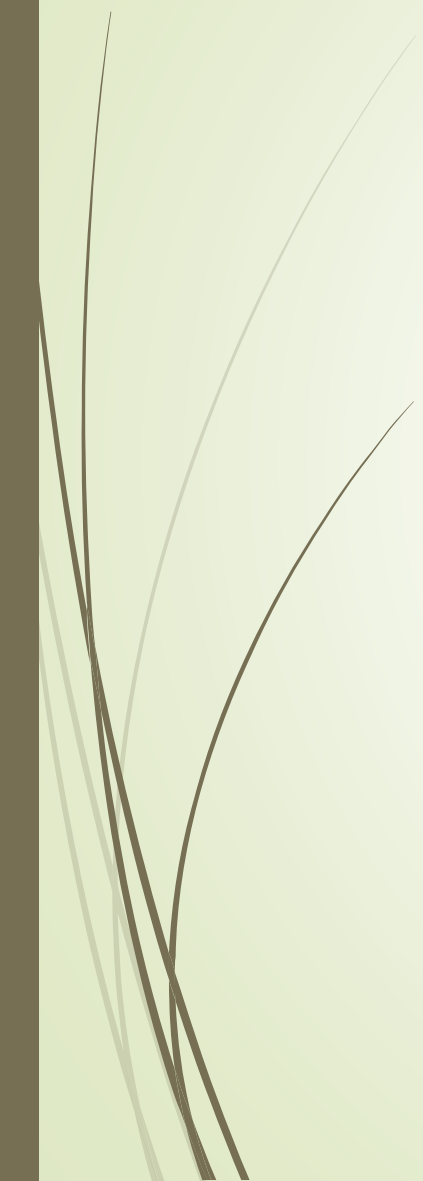
```
| private IEnumerator<bool> ChangePosition()  
| {  
|     while (true)  
|     {  
|         foreach (var primitive in _primitives)  
|         {  
|             var transform = Matrix.CreateRotationZ(0.001f);  
|             primitive.Position = Vector3.Transform(primitive.Position, transform);  
|         }  
|         yield return true;  
|     }  
| }
```

Iterator

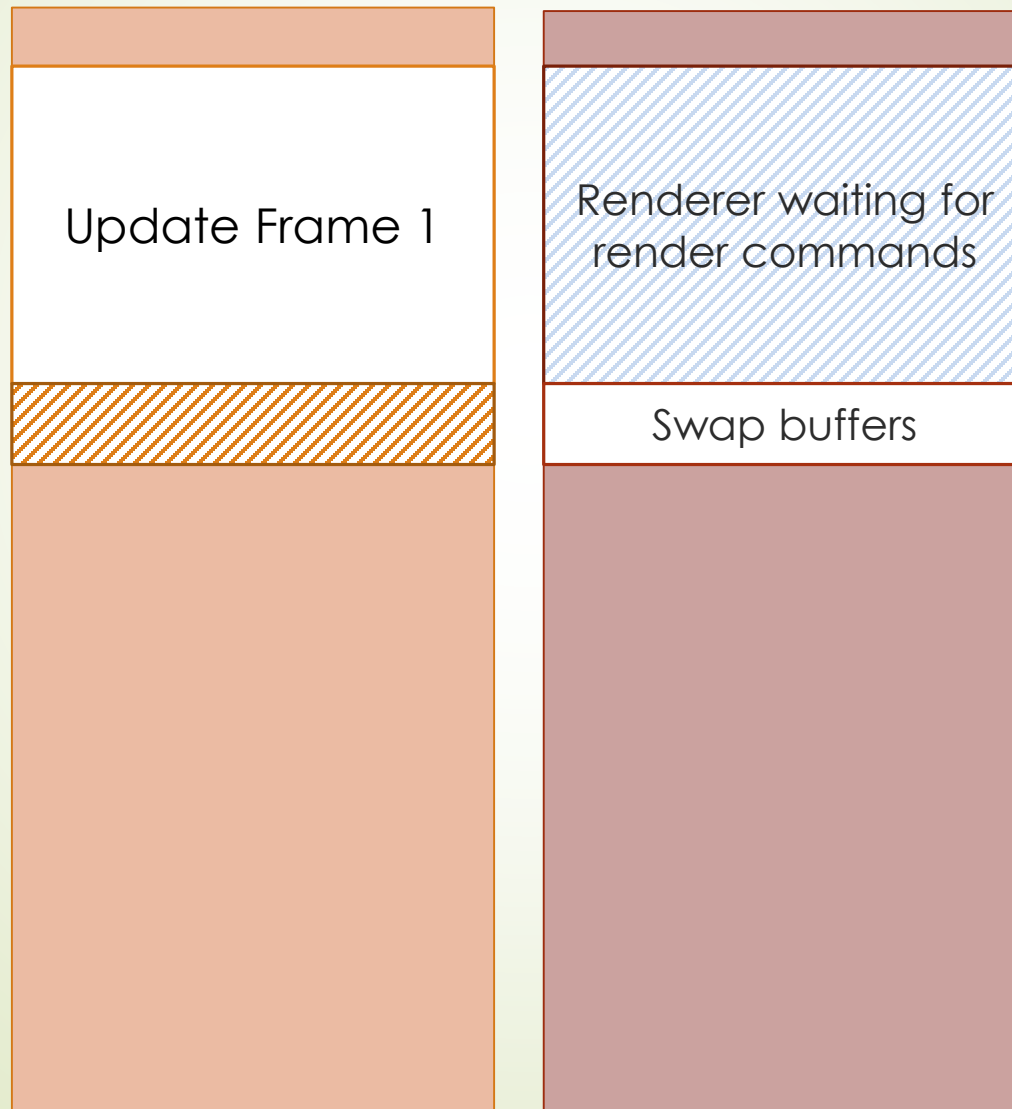
```
bool IEnumerator.MoveNext()  
{  
    switch (this.<>1__state)  
    {  
        case 0:  
            this.<>1__state = -1;  
            break;  
        case 1:  
            this.<>1__state = -1;  
            break;  
        case 2:  
            this.<>1__state = -1;  
            goto label_7;  
        default:  
            return false;  
    }  
    if (this.<>4__this._changeColourCoroutine.MoveNext())  
    {  
        this.<>2__current = 1;  
        this.<>1__state = 1;  
        return true;  
    }  
label_7:  
    if (this.<>4__this._changePositionCoroutine.MoveNext())  
    {  
        this.<>2__current = 1;  
        this.<>1__state = 2;  
        return true;  
    }  
    else  
        goto default;  
}  
  
[DebuggerHidden]  
void IEnumerator.Reset()  
{  
    throw new NotSupportedException();  
}  
  
void IDisposable.Dispose()  
{  
    }
```



Multi threaded rendering

- Double buffer
 - Triple buffer
 - Concurrent Collections
 - Jobs and workers
- 

Double Buffer



Wait for renderer to
switch buffers

Wait for renderer to
switch buffers

Wait for renderer to
switch buffers

Update Frame 1



Update Frame 2



Renderer waiting for
render commands



Swap buffers

Draw Frame 1



Renderer waiting for
render commands



Swap buffers

Draw Frame 2



Wait for renderer to
switch buffers

Wait for renderer to
switch buffers

Blocked until rendering
finished

Wait for renderer to
switch buffers

Update Frame 1

Update Frame 2

Update Frame 3

Renderer waiting for
render commands

Swap buffers

Draw Frame 1

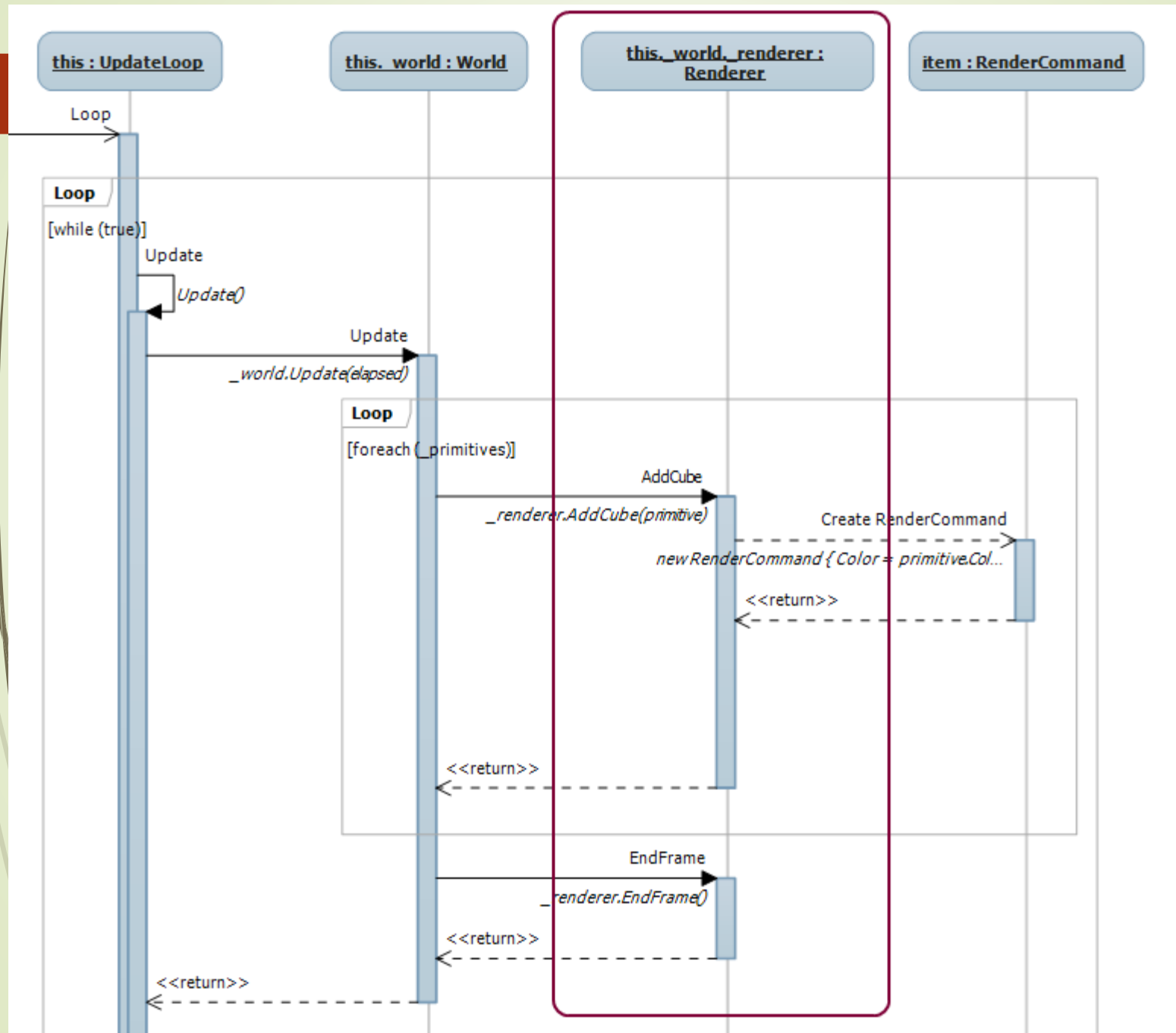
Renderer waiting for
render commands

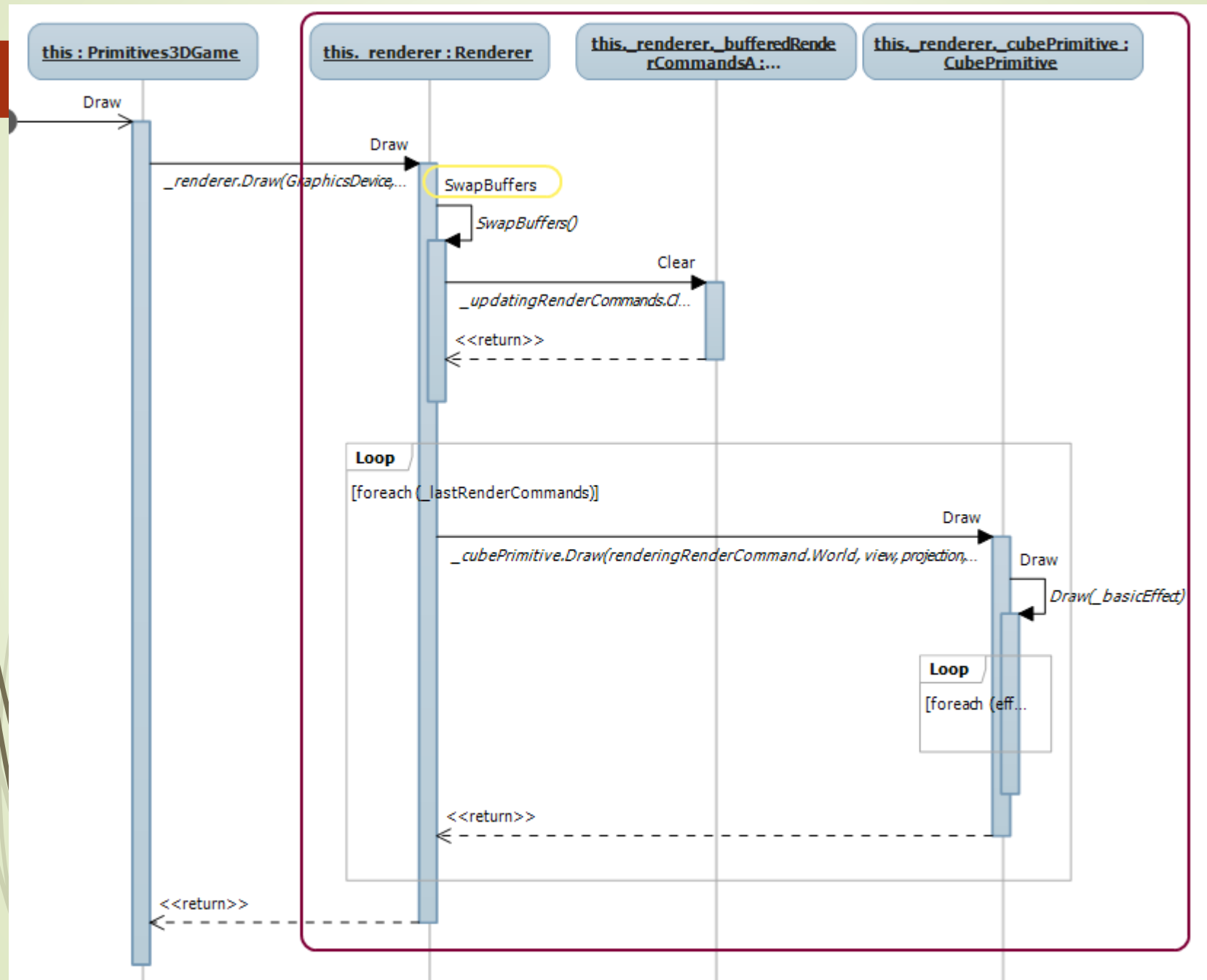
Swap buffers

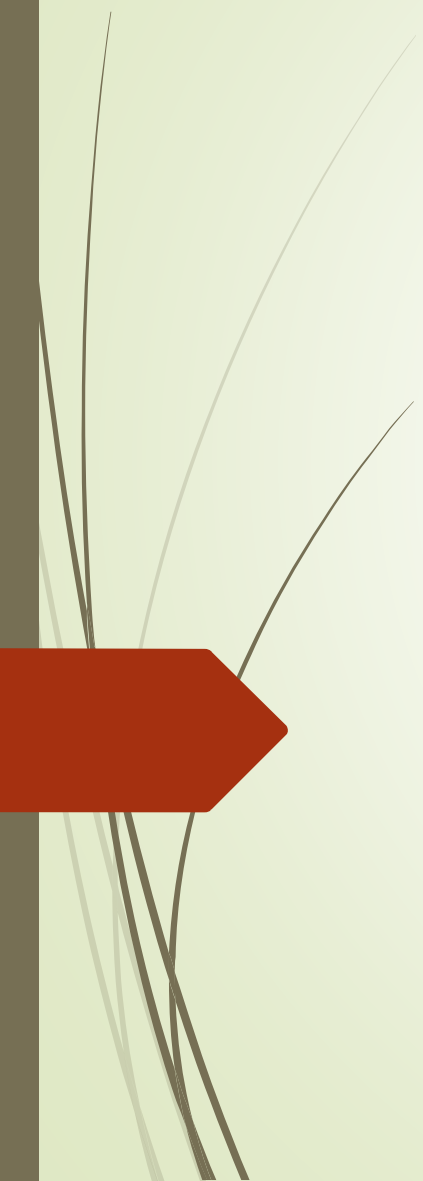
Draw Frame 2

Swap buffers

Draw Frame 3





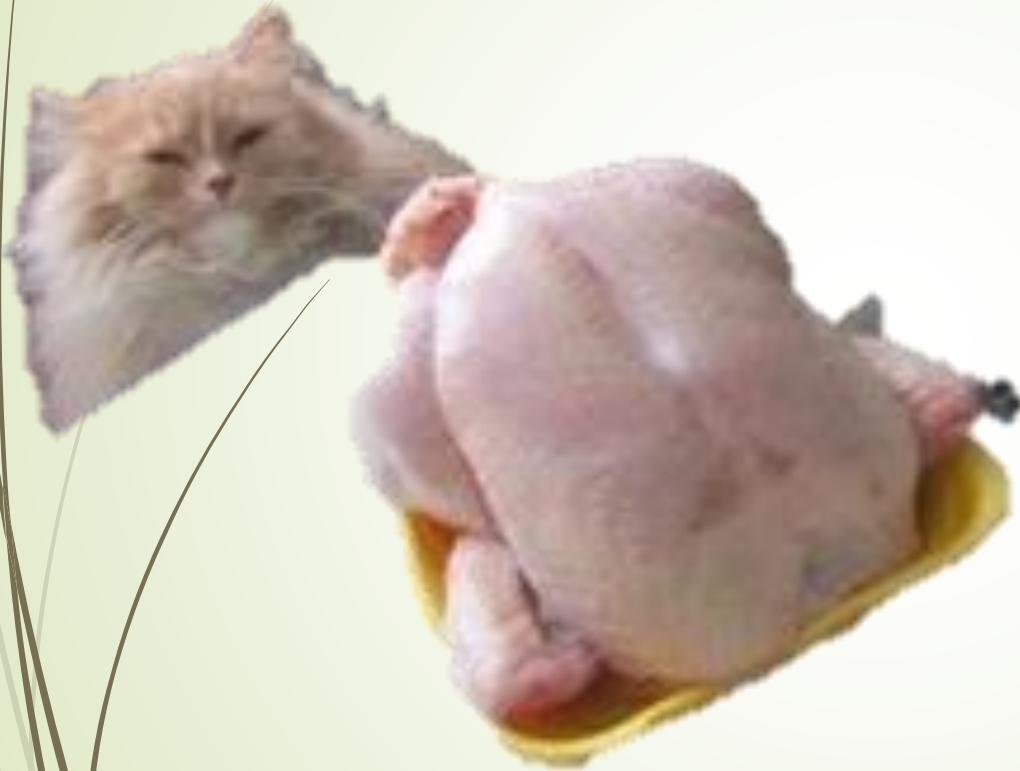


Why

Using concurrent Queue instead and Blocking Collection



Analysis



Comparision

Double buffer

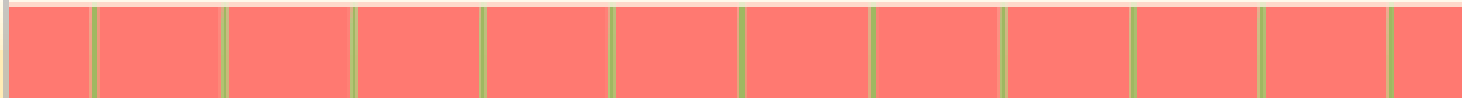
Main Thread(1308)

Worker Thread(4392)

Concurrent Queue

Main Thread(5696)

Worker Thread(3224)





Thanks

➤ More to read

<http://bitly.com/bundles/roundcrisis/3>

Andrea Magnorsky

@silverspoon

roundcrisis.com

batcatgames.com

